Module No. 1	Introduction
Computer network and its history, progress and application, Internet, Network architecture, Networking devices. OSI Model, TCP/IP Protocol stack, Networking in different OS.	

What is a Computer Network?

- A group of connected devices that can share information and resources.
- Devices can be computers, printers, and servers.
- Networks help send data quickly and easily.
  - ✓ Eg.,: Email, file sharing, and browsing the internet

Basic Terminologies of Computer Networks

- Network
- Nodes
- Protocol
- Topology
- Service Provider Networks
- IP Address
- Domain Name System (DNS)
- Firewall

Types of system

- **Open system**
- **Closed system**

Network Type

- LAN Local Area Network
- MAN Metropolitan Area Network
- WAN Wide Area Network
- PAN Personal Area Network
- BAN Body Area Network

Network Architecture

- Client-Server Architecture
- Peer-to-Peer Architecture

NETWORK DEVICES

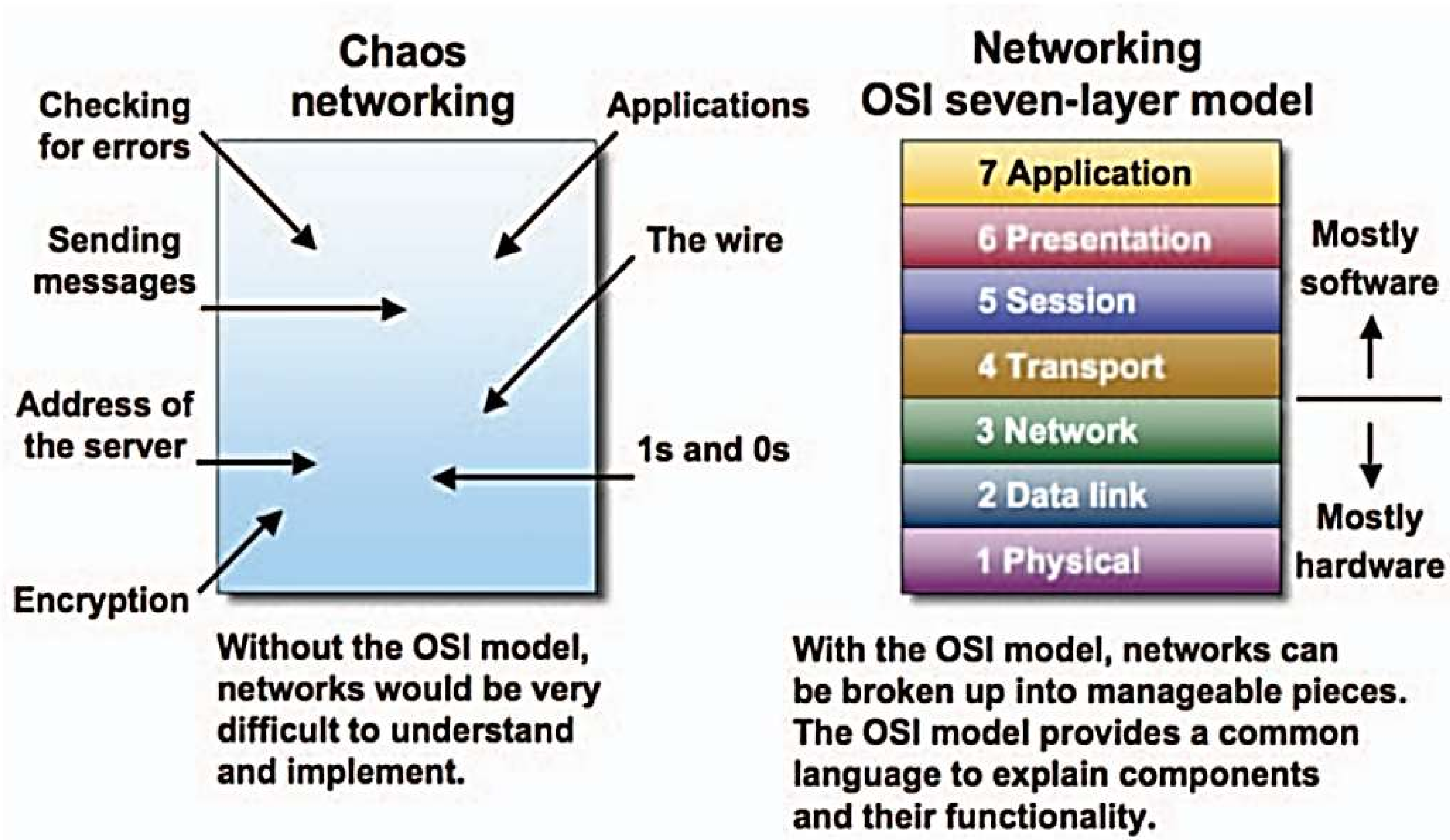
- **Modem**: Connects to the Internet by converting signals.
- **Ethernet Card**: Provides wired network connections.
- **RJ45**: Connector for Ethernet cables.
- **Repeater**: Extends network range by boosting signals.
- **Hub**: Basic device that broadcasts data to all connected devices.
- **Switch**: Efficiently directs data to specific devices.
- **Router**: Connects different networks and manages traffic.
- **Gateway**: Bridges networks with different protocols.
- **Wi-Fi Card**: Enables wireless network connectivity.

Layered Architecture

- Sometimes called Protocol Layering.
- When communication is simple, only one simple protocol is needed.
- When the communication is complex, the task is divided between different layers.
- Each layer is to be equipped with different protocol as per requirement.
- This is called “Protocol Layering”.

Two models exist:

- OSI model
- TCP/IP model


Chaos networking
Checking for errors
Sending messages
Address of the server
Encryption
Applications
The wire
1s and 0s
Networking OSI seven-layer model
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data link
1 Physical
Mostly software
Mostly hardware
Without the OSI model, networks would be very difficult to understand and implement.
With the OSI model, networks can be broken up into manageable pieces. The OSI model provides a common language to explain components and their functionality.

Introduction to the OSI Model

- OSI stands for **Open Systems Interconnection**
- Created by International Standards Organization
- Was created as a framework and reference model to explain how different networking technologies work together and interact
- Is not a standard that networking protocols must follow
- Each layer has specific functions it is responsible for
- All layers work together in the correct order to move data around a network

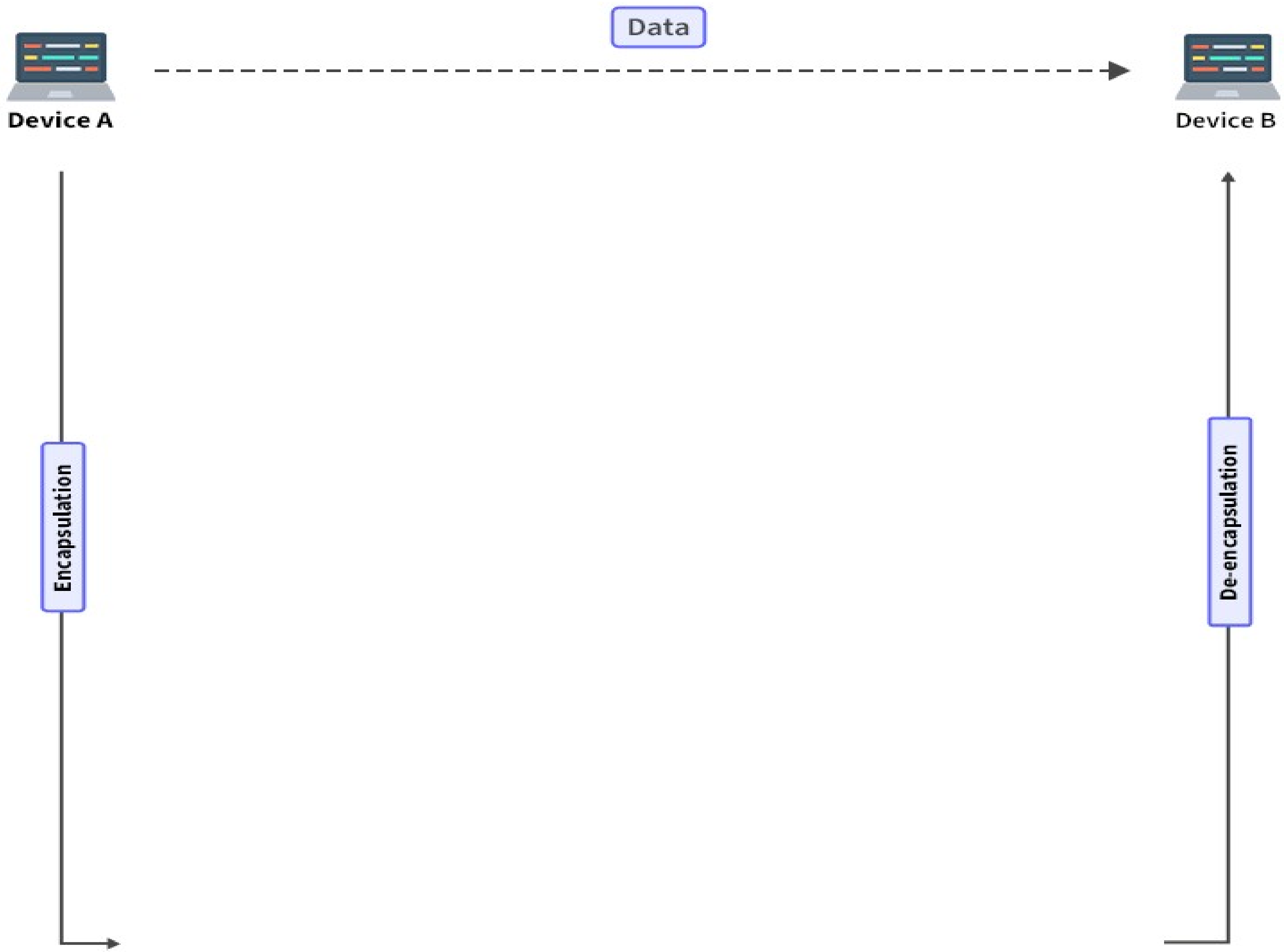
How Data Is Referred to in the OSI Model

Application layer	Data
Presentation layer	
Session layer	
Transport layer	Segment
Networking layer	Packet
Data Link layer	Frame
Physical layer	Bits

The process of moving data between layers of the OSI Model

Encapsulation: Data > Segment > Packet > Frame > Bits

De-encapsulation: Bits > Frame > Packet > Segment > Data



* PH : Presentation Header

OSI Model

- **07 Application** — The closest layer to the user; provides application services.
- **06 Presentation** — Encrypts, encodes and compresses usable data.
- **05 Session** — Establishes, manages, and terminates sessions between end nodes.
- **04 Transport** — Transmits data using transmission protocols including TCP & UDP.
- **03 Network** — Assigns global addresses to interfaces and determines the best routes through different networks.
- **02 Data link** — Assigns local addresses to interfaces, delivers information locally, MAC method
- **01 Physical** — Encodes signals, cabling and connectors, physical specifications.

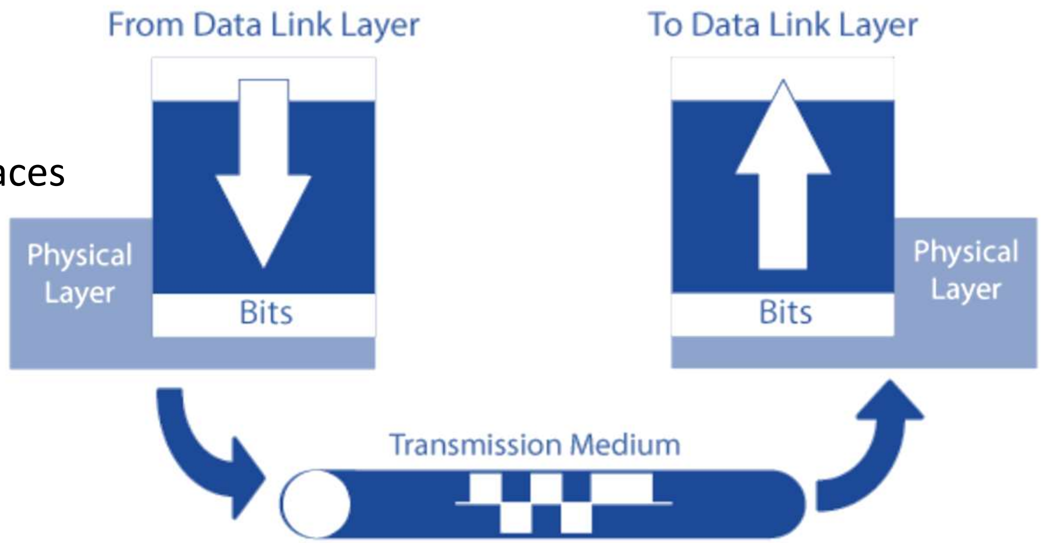
Layer 1 – The Physical Layer

- It is the first and lowest layer of the OSI model.
- It deals with the actual physical connection between devices (computers, routers, etc.).
- It is responsible for sending and receiving raw data (bits) like 0s and 1s.
- This layer does not understand the data — it only moves it.
- It uses wires, cables, switches, and other hardware to send data.
- Defines how signals (electric or light) are used to represent data.
- It decides how fast the data is sent (data transmission speed). It represents the **bit rate or bandwidth capacity** of the communication channel — how many bits can be transmitted per second.
- Examples of physical layer devices:
  - ✓ Cables (like LAN cables), Network Interface Card (NIC), Hubs and Switches.

Analogy:

Think of it as the road or path where data travels — it doesn't care what's inside the car (data), it just makes sure the car moves.

- Components of the physical layer include:
  - Cabling system components
  - Adapters that connect media to physical interfaces
  - Connector design and pin assignments
  - Hub, repeater, and patch panel specifications
  - Wireless system components
  - Network Interface Card (NIC)



- ✓ **Bit rate**: Number of bits (0s and 1s) sent per second (e.g., 100 Mbps).
- ✓ **Bandwidth**: The maximum data-carrying capacity of the medium (cable or wireless).

Layer 2 – The Data Layer

- It is the 2nd layer of the OSI model (just above the Physical Layer).
- It is responsible for node-to-node communication (between two directly connected devices).
- It organizes **data into frames** (small blocks of data) for easier transmission.
- Adds **MAC addresses** (hardware addresses) to identify the sender and receiver on a local network.
- Provides **error detection** using techniques like CRC (Cyclic Redundancy Check).
- Performs **flow control** to avoid overwhelming the receiver.
- Ensures reliable delivery over a single physical link (e.g., between your computer and a switch).
- Detects and may correct basic transmission errors (from electrical noise, etc.).
- Works only between devices on the same network segment (LAN).
- Example protocols: Ethernet, PPP (Point-to-Point Protocol), HDLC

- Common networking components
  - ✓ Network interface cards - Have a layer 2 or MAC address
  - ✓ Ethernet and Token Ring switches - A switch uses the MAC address to filter and forward traffic, helping relieve congestion and collisions on a network segment.
  - ✓ Bridges

    Find out: Switch Vs Bridge

Analogy:

It's like a packaging department:

*It **wraps data into frames**, labels them with MAC addresses, checks for errors, and controls how fast the packages are sent to avoid overload.*

Layer 3 – The Network Layer

- It is the 3rd layer of the OSI model.
- Responsible for moving data between different networks (not just devices on the same LAN).
- **Handles routing** – choosing the best path for data to travel from source to destination.
- **Uses IP addresses** to identify devices across networks.
- Breaks data into packets and reassembles them at the receiver's end.
- Deals with **logical addressing** (like IP addresses), not hardware addresses.
- Can detect and avoid network congestion (too much traffic).
- Supports both connectionless and connection-oriented delivery.
- Communicates with **routers**, which direct packets between networks.
- Example protocols: IP (Internet Protocol), ICMP, IPsec

- Provides an end-to-end logical addressing system so that a packet of data can be routed across several layer 2 networks (Ethernet, Token Ring, Frame Relay, etc.).
- Network layer addresses can also be referred to as logical addresses.

The Internet Protocol (IP) addresses make networks easier to both set up and connect with one another.

- To make it easier to manage the network and control the flow of packets, many organizations separate their network layer addressing into smaller parts known as subnets. Routers use the network or subnet portion of the IP addressing to route traffic between different networks.
- Some basic security functionality can also be set up by filtering traffic using layer 3 addressing on routers or other similar devices.

Analogy:

It's like a GPS : It chooses the best path for sending data across cities (networks), using IP addresses like street addresses.

Layer 4 – The Transport Layer

- Offers end-to-end communication between end devices through a network.
- Depending on the application, the transport layer either offers reliable, connection-oriented or connectionless, best-effort communications.
- Some of the functions offered by the transport layer include:
  - ❖ Confirmation that the entire message arrived intact
  - ❖ Segmentation of data for network transport
  - ❖ Control of data flow to prevent memory overruns
  - ❖ Establishment and maintenance of both ends of virtual circuits
  - ❖ Transmission-error detection
  - ❖ Realignment of segmented data in the correct order on the receiving side
  - ❖ Multiplexing or sharing of multiple sessions over a single physical link
- The most common transport layer protocols are the connection-oriented TCP Transmission Control Protocol (TCP) and the connectionless UDP User Datagram Protocol (UDP).

Layer 5 – The Session Layer

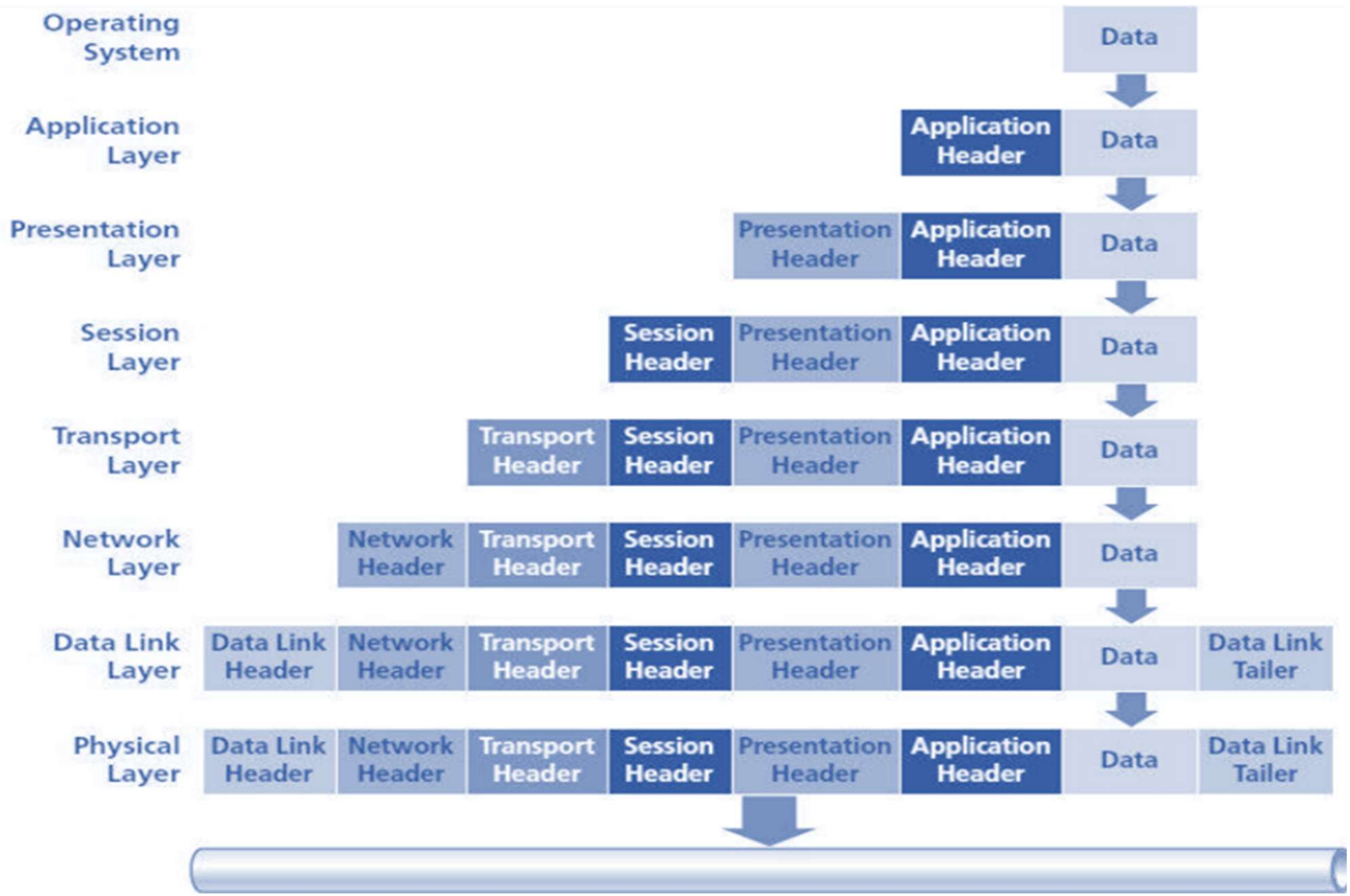
- Provides various services, including tracking the number of bytes that each end of the session has acknowledged receiving from the other end of the session.
- Allows applications functioning on devices to establish, manage, and terminate a dialog through a network.
- Session layer functionality includes:
    - ❖ Virtual connection between application entities
    - ❖ Synchronization of data flow
    - ❖ Creation of dialog units
    - ❖ Connection parameter negotiations
    - ❖ Partitioning of services into functional groups
    - ❖ Acknowledgements of data received during a session
    - ❖ Retransmission of data if it is not received by a device

Layer 6 – The Presentation Layer

- Responsible for how an application formats the data to be sent out onto the network.
- Allows an application to read (or understand) the message.
- Some functionalities of Presentation layer includes:
  - ❖ Encryption and decryption of a message for security
  - ❖ Compression and expansion of a message so that it travels efficiently
  - ❖ Graphics formatting
  - ❖ Content translation
  - ❖ System-specific translation

Layer 7 – The Application Layer

- Provides an interface for the end user operating a device connected to a network.
- This layer is what the user sees, in terms of loading an application (such as Web browser or e-mail); that is, this application layer is the data the user views while using these applications.
- Some functionalities of Application layer includes:
  - ❖ Support for file transfers
  - ❖ Ability to print on a network
  - ❖ Electronic mail
  - ❖ Electronic messaging
  - ❖ Browsing the World Wide Web


Operating System
Data
Application Layer
Application Header
Data
Presentation Layer
Presentation Header
Application Header
Data
Session Layer
Session Header
Presentation Header
Application Header
Data
Transport Layer
Transport Header
Session Header
Presentation Header
Application Header
Data
Network Layer
Network Header
Transport Header
Session Header
Presentation Header
Application Header
Data
Data Link Layer
Data Link Header
Network Header
Transport Header
Session Header
Presentation Header
Application Header
Data
Data Link Tailer
Physical Layer
Data Link Header
Network Header
Transport Header
Session Header
Presentation Header
Application Header
Data
Data Link Tailer

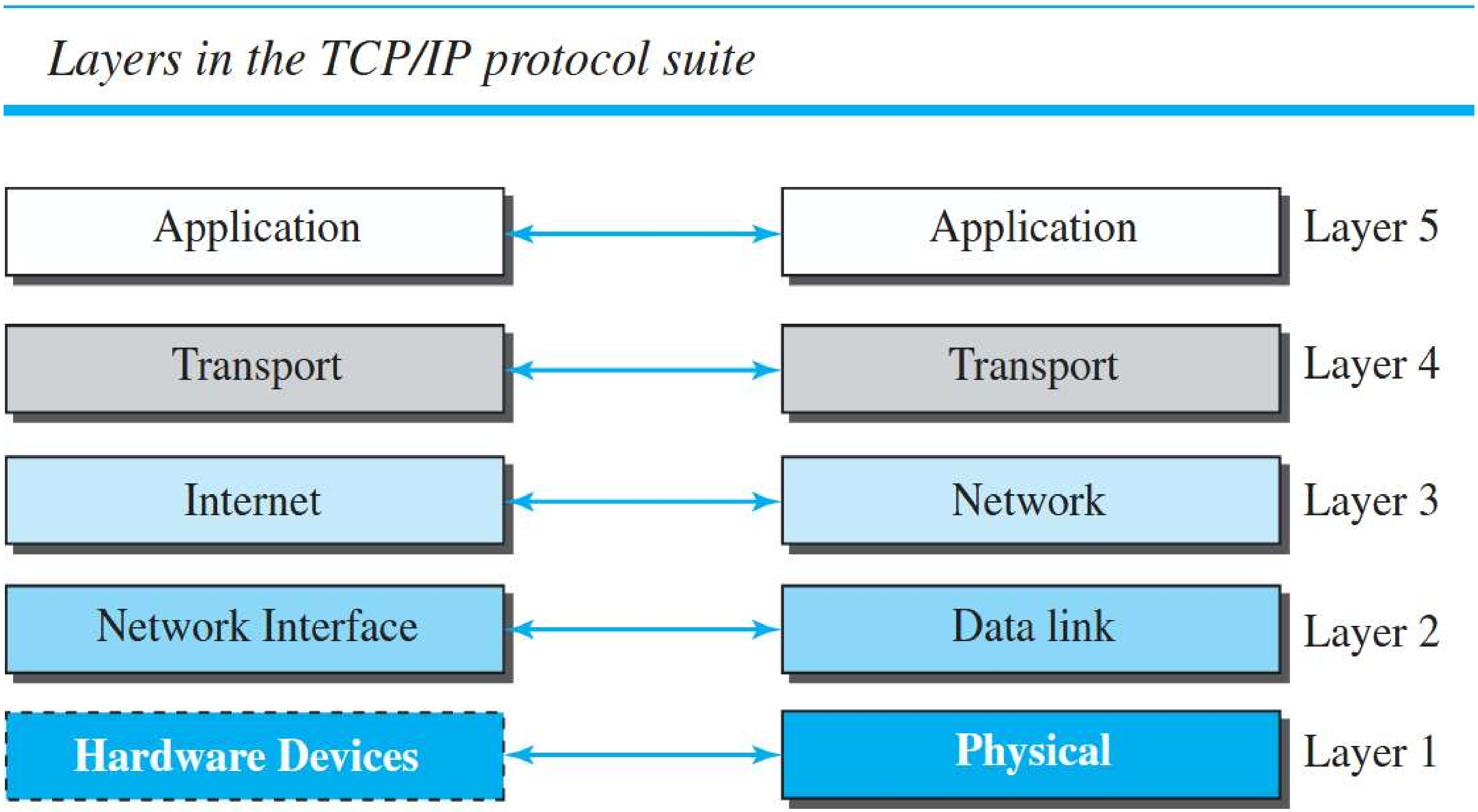
- Each layer of OSI Model except Physical layer adds its own header to the data that originated from the operating system
  - ✓ Adds own header in front of the header from the previous layer
  - ✓ Header contains information that describes what each layer of the OSI Model should do with the data
- Data Link layer also adds a tailer
  - ✓ Tailer contains additional information that deals with error correction

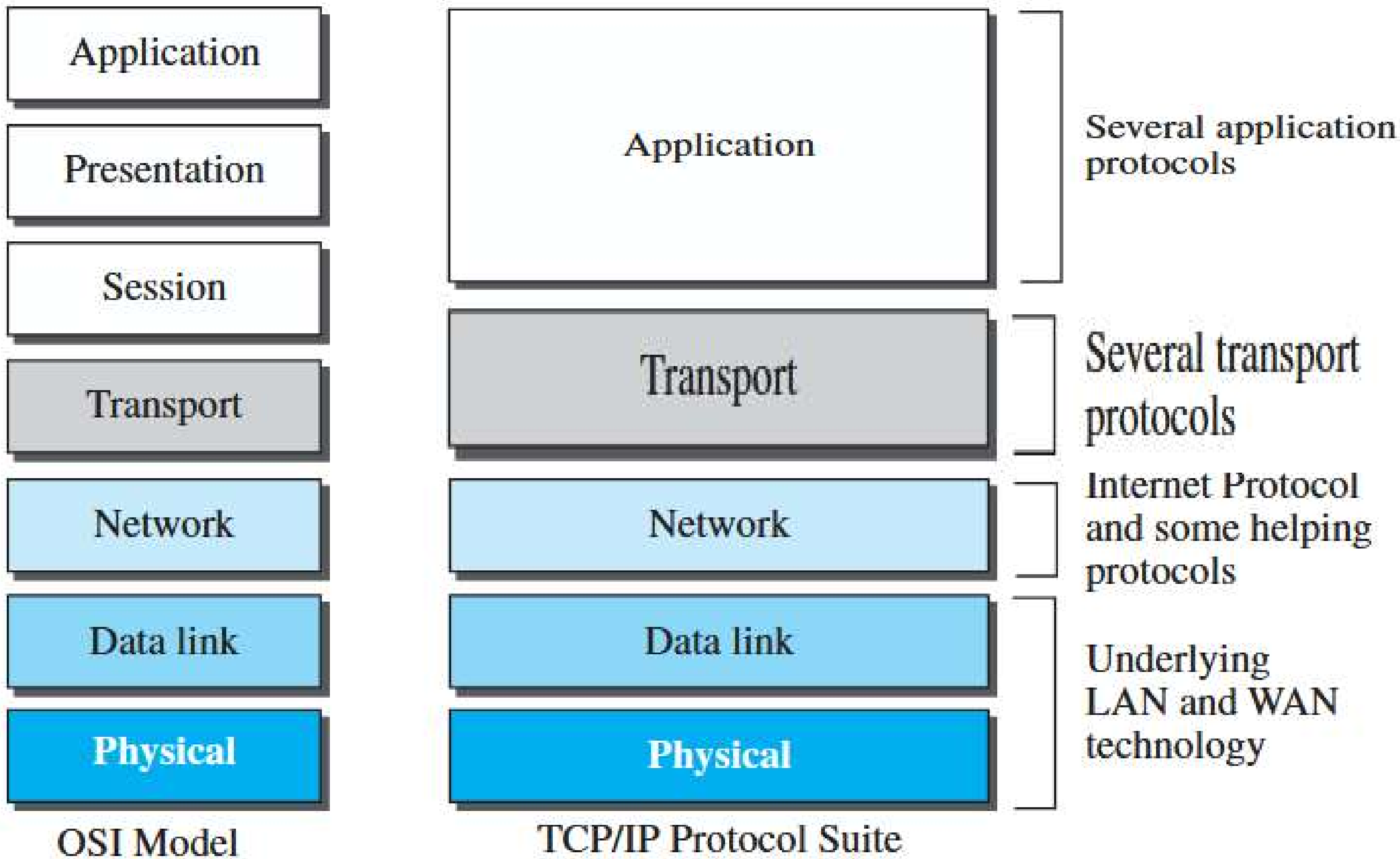
A message of 3000 bytes is generated by an application and transmitted over a network. As the data moves through the OSI layers, each of the following headers is added: 20 bytes by the Transport layer, 20 bytes by the Network layer, 18 bytes by the Data Link layer, and 8 bytes of framing overhead at the Physical layer.

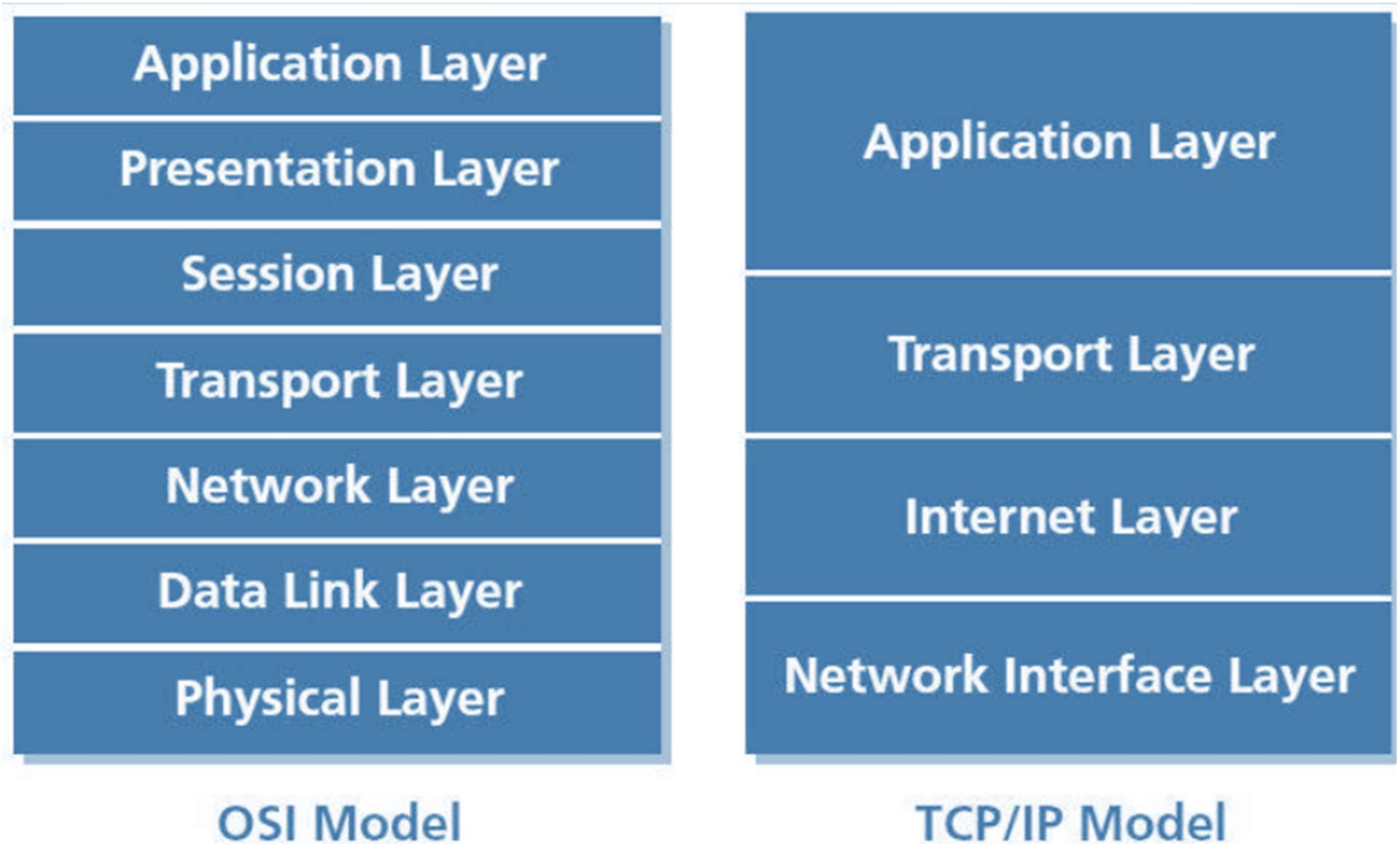
Lack of OSI Model's Success

- OSI was completed when TCP/IP was fully in place and a lot of time and money had been spent on the suite; changing it would cost a lot.
- Some layers in the OSI model were never fully defined.

TCP/IP model or Protocol Suite

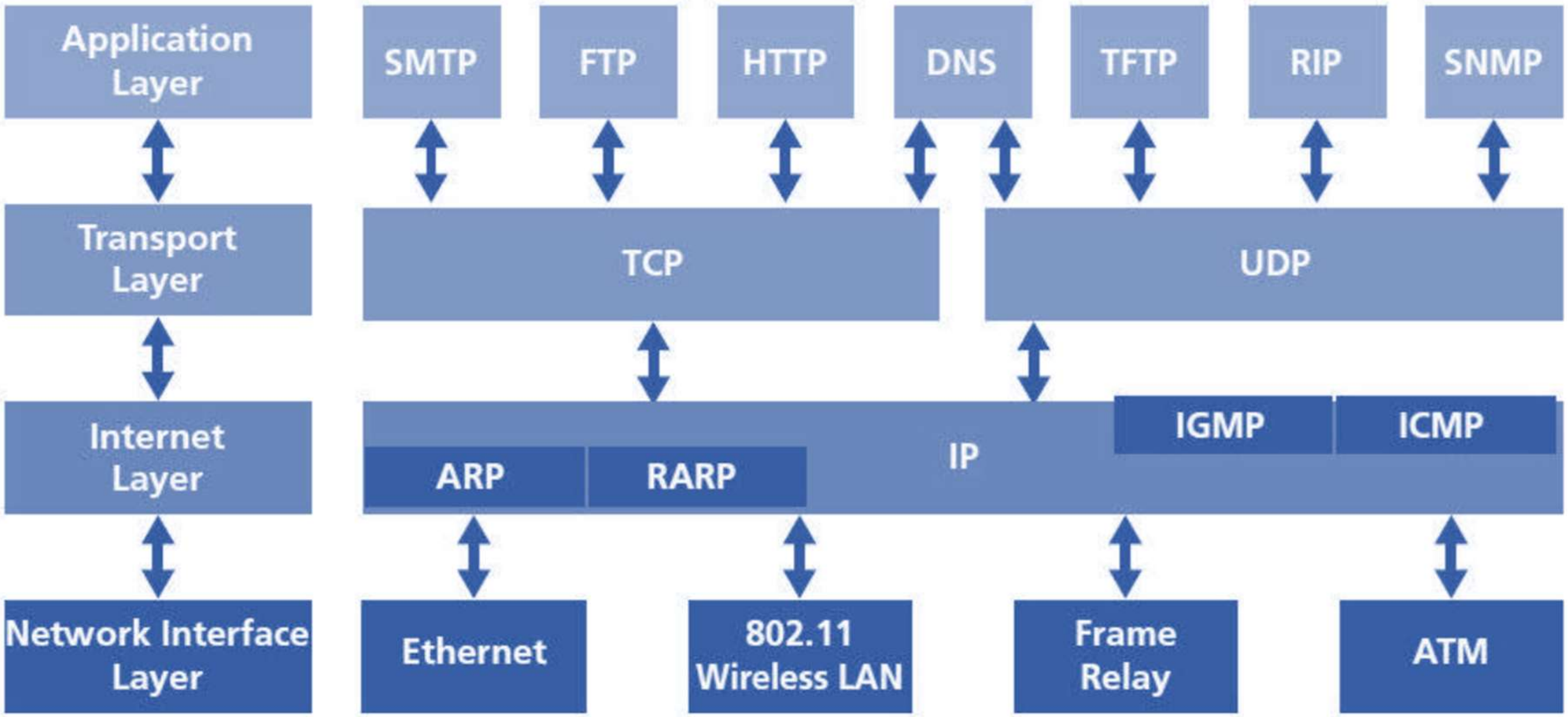


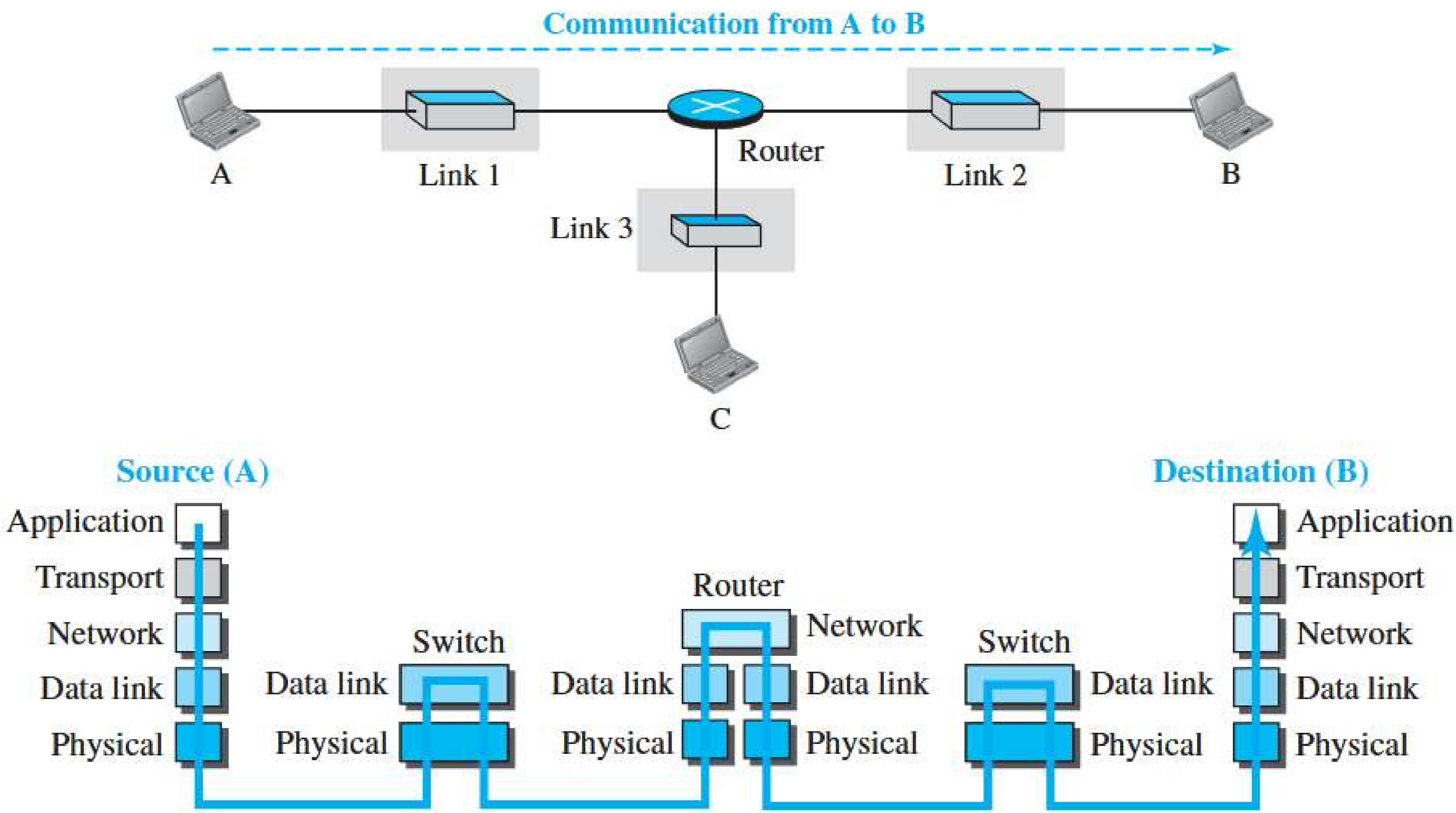

Application
Presentation
Session
Transport
Network
Data link
Physical
OSI Model
Application
Transport
Network
Data link
Physical
TCP/IP Protocol Suite
Several application protocols
Several transport protocols
Internet Protocol and some helping protocols
Underlying LAN and WAN technology


Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
OSI Model
Application Layer
Transport Layer
Internet Layer
Network Interface Layer
TCP/IP Model

Built around the **TCP/IP protocol suite**

– A protocol suite is a large number of related protocols that work together to allow networked computers to communicate.





Communication from A to B
A
Link 1
Router
Link 2
B
Link 3
C
Source (A)
Application
Transport
Network
Data link
Physical
Switch
Data link
Physical
Router
Data link
Physical
Network
Data link
Physical
Switch
Data link
Physical
Destination (B)
Application
Transport
Network
Data link
Physical

Addressing in the TCP/IP protocol suite

Packet names	Layers	Addresses
Message	Application layer	Names
Segment / User datagram	Transport layer	Port numbers
Datagram	Network layer	Logical addresses
Frame	Data-link layer	Link-layer addresses
Bits	Physical layer	

Topics to be read by students:

- History of Computer Networks, Current trend
- Where can be this Computer Networks is implemented and it’s usage
- Networking in Windows OS, Linux OS